

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION II
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

INDEX NUMBER

BIOLOGY

9648/02

Paper 2 Core Paper

17 September 2013

2 hours

Additional Materials: Answer Paper
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **any one** question.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	10
2	14
3	16
4	14
5	15
6	11
Section B	
7 / 8	20
Total	100
Percentage	
Grade	

This document consists of **18** printed pages.



Section A
Answer **all** questions.

- 1 Fig. 1 is an electron micrograph of a cell found in the pancreas.

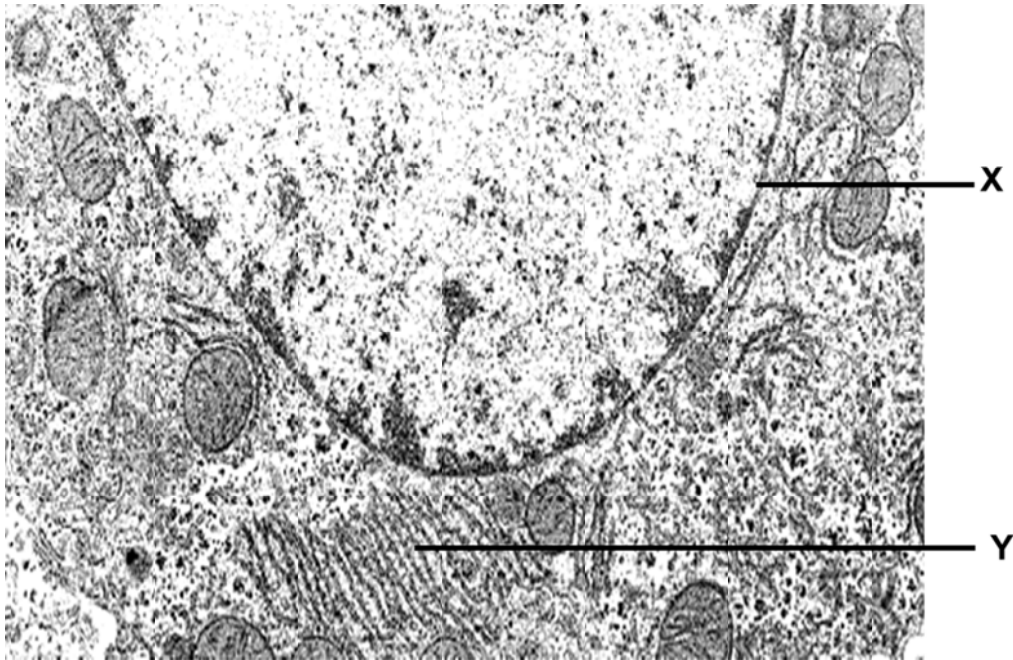


Fig. 1

- (a) Describe how organelle X is adapted for genetic expression.

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- (b) Describe how insulin, after being synthesized by ribosomes on Y, is exported out of the cell.

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- (c) When insulin is released into the bloodstream, it exerts many effects on carbohydrate, lipid and protein metabolism.

One of its effects is to activate the enzyme glycogen synthase, which functions to polymerize glucose monomers into glycogen.

- (i) Explain how the structure of glycogen enables it to perform its function in animals.

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- (ii) Describe **two** ways in which the structure of glycogen differs from the structural polysaccharide found in plant cell walls.

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[Total: 10]

- 2 Fig. 2.1 shows the structure of a protease enzyme found in a retrovirus.

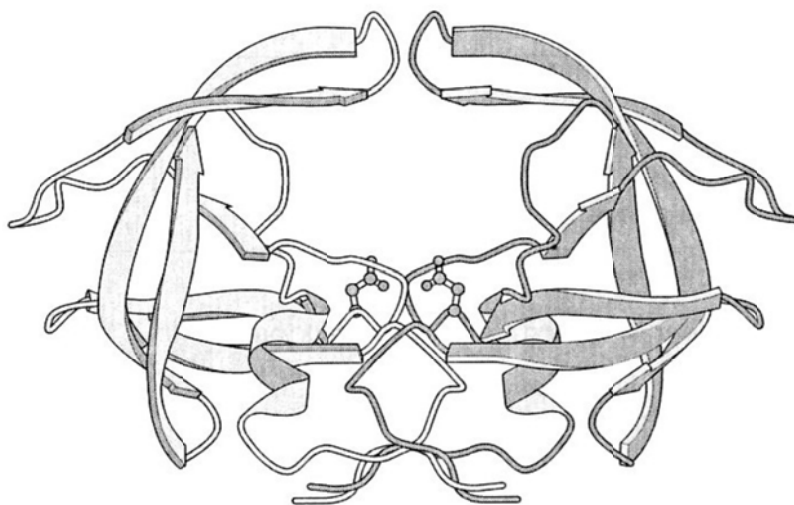


Fig 2.1

- (a) (i) With reference to Fig. 2.1, state the evidences that the enzyme has both a tertiary and quaternary protein structure.

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- (ii) Explain what determines the 3-dimensional configuration of the enzyme.

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- (b) The rate of reaction of HIV protease may be decreased by different inhibitors.

Fig. 2.2 shows the effect of substrate concentration on the initial rate of an enzyme catalysed reaction with and without the addition of a non-competitive inhibitor.

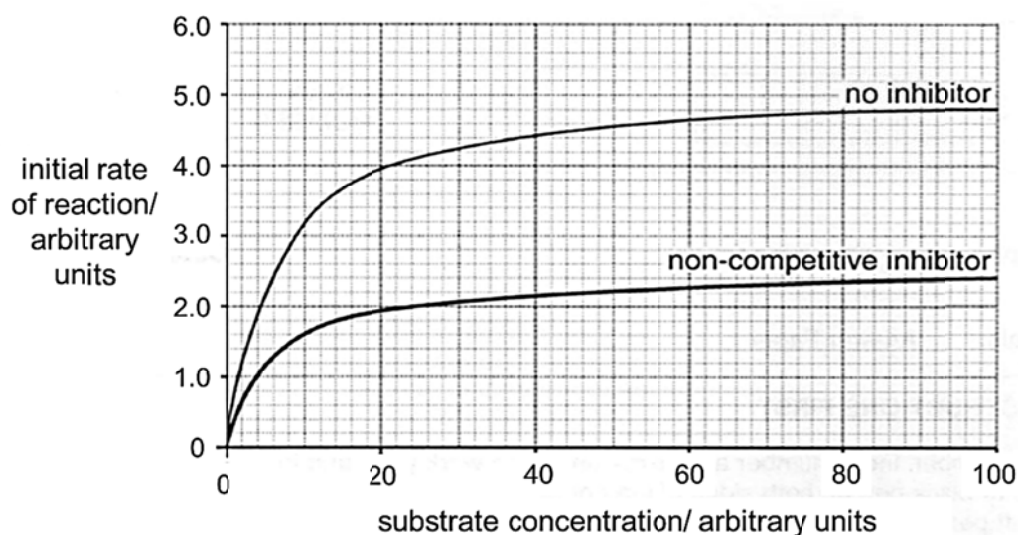


Fig 2.2

- (i) On Fig. 2.2 draw and label the curve you would expect in the presence of a competitive inhibitor. [1]

- (ii) Describe and explain the difference in the shape of the curve you have drawn and that of the non-competitive inhibitor.

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- (c) A similar protease is found in blackflies. A proteinaceous toxin found in insecticides used to control its population can be activated by this protease.

The active toxin binds to receptor molecules on the larvae's gut epithelial cell membranes. This damages the epithelial cells and kills the larvae.

Fig. 2.3 shows the results of an experiment to determine the optimum pH for activation of the toxin.

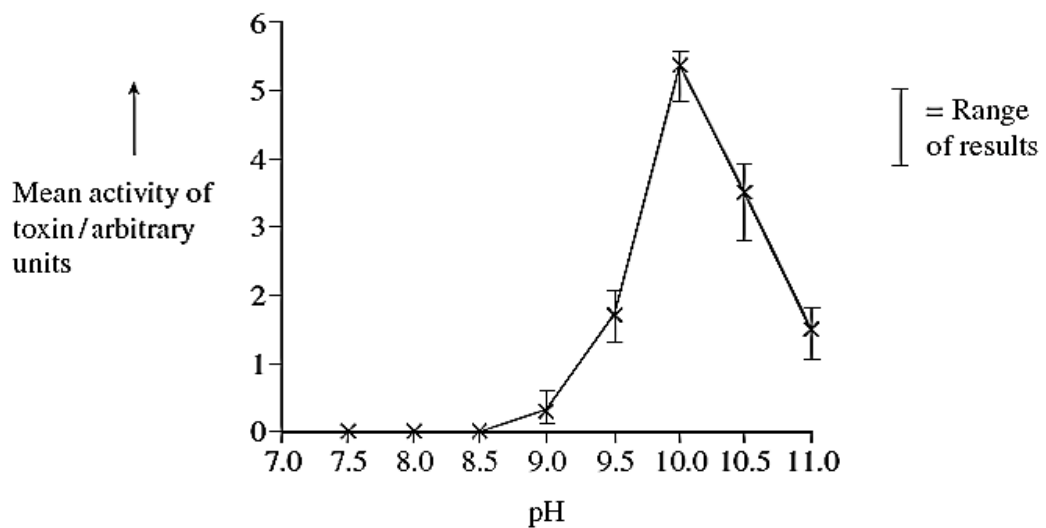


Fig. 2.3

- (i) With reference to Fig. 2.3, describe the effect of pH on mean activity of the toxin.

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- (ii) Explain why there is no activity of toxin when pH falls below 8.5.

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- (iii) Suggest how the protease in larva's digestive system causes the toxin to be activated.

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[Total: 14]

- 3 Fig. 3.1 shows a T4 bacteriophage which has been modified to contain the DNA of Mu bacteriophage, a known temperate phage.

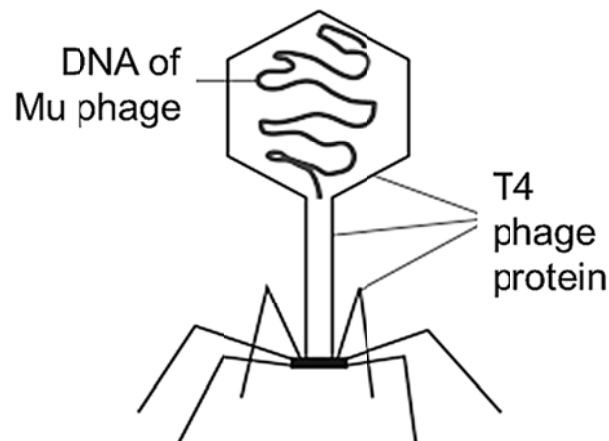


Fig. 3.1

- (a) Suggest how this bacteriophage attaches to the correct host cell.

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- (b) Outline the main events which occur following the entry of the phage DNA into the host cell until new phage particles are released.

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While the T4 bacteriophage infects bacterial cells, the influenza virus infects cells of the respiratory tract in animals.

H1N1 influenza virus causes swine flu in pigs. In a number of instances, people have developed the swine flu infection when they are closely associated with pigs, for example farmers and pork processors.

Figure 3.2 shows a H1N1 virus.

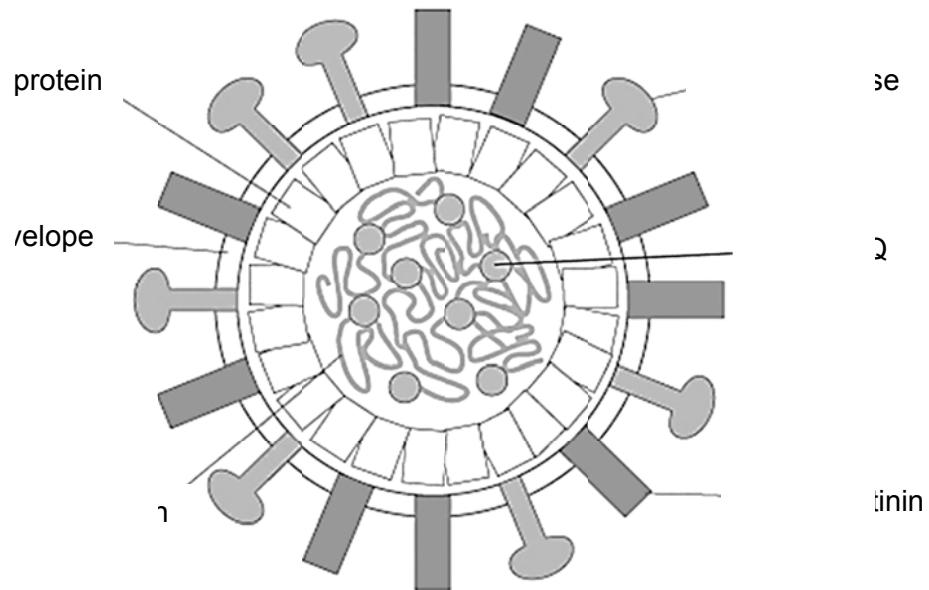


Fig. 3.2

- (c) Describe the significance of enzyme Q to the reproductive cycle of the H1N1 virus.

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- (d) A person is infected with the H1N1 influenza virus but does not develop the symptoms. Examination shows that the person's immune system has produced an antibody that binds to the haemagglutinin molecules.

- (i) Suggest why the person does not develop influenza-like symptoms.

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- (ii) After a few days and many viral generations, the person begins to develop influenza symptoms. Tests for the antibody referred to above show that it is still present in the blood.

Account for the observations and comment on the implication this has for developing H1N1 influenza vaccines.

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- (e) Contrast the reproductive cycles of the H1N1 influenza virus and Human Immunodeficiency Virus (HIV).

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[Total: 16]

- 4 Oxytocin (OT) is a neuropeptide secreted from the posterior pituitary gland in the brain and is best known for its role in mammalian birth and lactation.

In response to a variety of stimuli such as suckling, parturition, or certain kinds of stress, the processed OT peptide is released from the posterior pituitary into the systemic circulation.

The OT receptor's activity is mediated by G proteins which activate a phosphatidylinositol-calcium second messenger system.

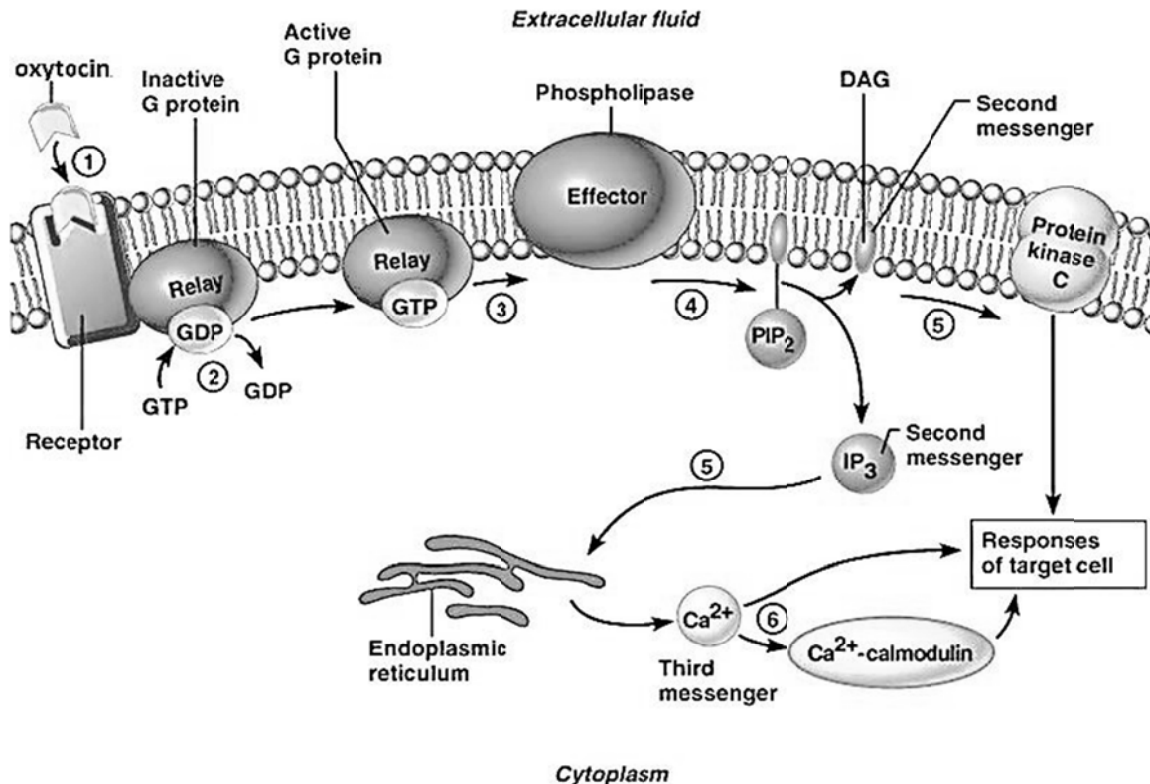


Fig. 4.1

- (a) The blood brain barrier is set by the endothelial cells lining the vessels in the brain.

Suggest why these cells can prevent oxytocin secreted from the gland to re-enter the brain.

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- (b) With reference to Fig. 4.1, describe and explain how a molecule of oxytocin can lead to signal amplification leading to cellular responses.

[5]

γ -Aminobutyric acid (GABA) is the chief inhibitory neurotransmitter in the mammalian central nervous system involving mature neurons. During foetal development, the release of GABA leads to an excitatory post-synaptic potential in immature neurons.

During labour, findings shown that high levels of maternal oxytocin induced a switch in the action of GABA from excitatory to inhibitory in the neurons of the foetus at birth hence increases their resistance to damage during delivery.

Fig. 4.2 shows the generation of the post-synaptic potentials in immature and mature neurons.

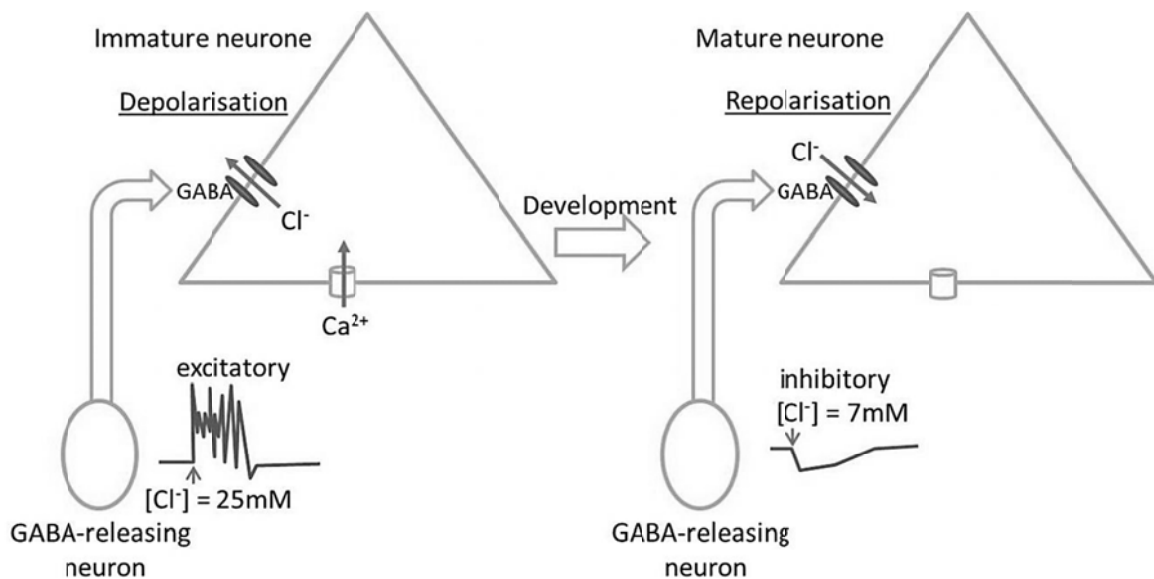


Fig. 4.2

- (c) Describe and explain how GABA is released from the pre-synaptic neurone to the post-synaptic neurone to result in an inhibitory post-synaptic potential.

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- (d) With reference to Fig. 4.2, explain how GABA could have differing effects on the post-synaptic membrane in different neurons.

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[Total: 14]

- 5 Nail-patella syndrome (NPS) is a genetic disorder that results in small, poorly developed nails and kneecaps, but can also affect many other areas of the body, such as the elbows, chest, and hips. The genetic linkage between the NPS and ABO blood group loci was the third to be identified in humans.

The NPS gene codes for a transcription factor and a gain-of-function mutation in one of its alleles leads to NPS.

An unaffected man with blood group O marries an affected female with blood group B. Fig. 5 shows the resultant pedigree.

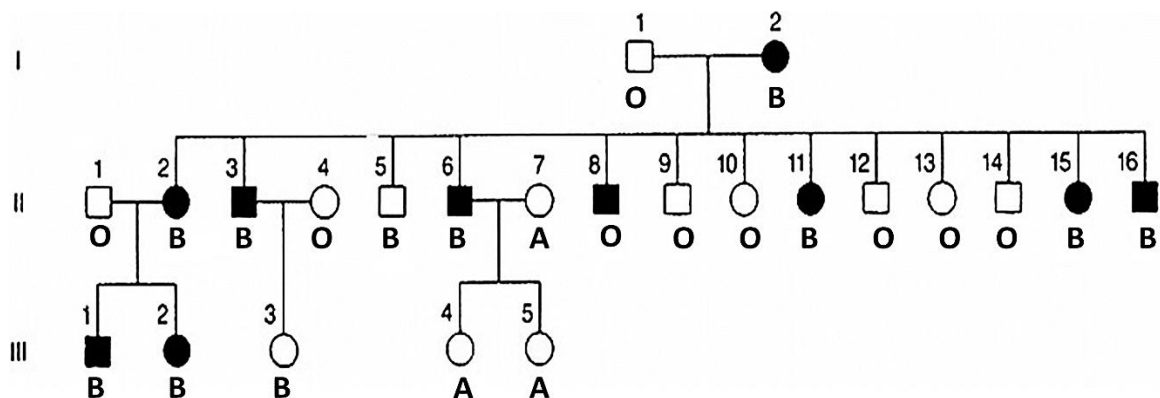


Fig. 5

(a) With reference to Fig. 5,

- (i) State and explain the mode of inheritance of NPS with respect to the ABO locus.

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- (ii) Explain why the mating between individual 6 and 7 in generation II produces 100% unaffected offsprings.

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- (b) (i) State the genotype of the individuals 5 and 8 in Generation II with respect to both the NPS and ABO loci.

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(ii) Suggest reasons for your answers in (i).

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(c) Apart from the factors due to the inheritance of genes, environment plays a very important role in determining phenotype of certain organisms.

Explain, with one example, how the environment may affect the phenotype.

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[Total: 15]

- 6 Cranes are large birds and one of the endangered species, *Grus americana* commonly known as the Whooping crane is the tallest North American bird which faced near extinction due to unregulated hunting and loss of habitat.

Cranes can be categorised based on their behaviours and one of the earliest methods of classifying cranes was based on the calls they make during the breeding season.

- (a) Suggest why biologists could use calls to investigate relationships between different species of crane.

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More recently, biologists have used DNA hybridisation to confirm the relationships between different species of crane. They made samples of hybrid DNA from the same and from different species. They measured the percentage of hybridisation of each sample. The results are shown in Table 6.

Table 6

Species of crane from which hybrid DNA was made	Percentage DNA hybridisation/%
<i>Grus americana</i> and <i>Grus monachus</i>	97.4
<i>Grus monachus</i> and <i>Grus rubicunda</i>	95.7
<i>Grus americana</i> and <i>Grus rubicunda</i>	35.5
<i>Grus rubicunda</i> and <i>Grus rubicunda</i>	99.9
<i>Grus americana</i> and <i>Grus americana</i>	99.9
<i>Grus monachus</i> and <i>Grus monachus</i>	99.8

Blank lined paper for writing.

[2]

[2]

[Turn over

Section B
Answer **one** question

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

- 7 **(a)** Describe the eukaryotic processing of pre-mRNA in terms of intron splicing, polyadenylation and 5'capping. [8]
- (b)** Define control elements and explain how they influence transcription in the eukaryotic genome. [6]
- (c)** State the various ways in which gene expression of an enzyme may be controlled at translational and post-translational level. [6]

[Total: 20]

- 8 **(a)** Describe the role of mitosis in maintaining genetic stability. [4]
- (b)** Explain how meiosis can lead to genetic variation. [8]
- (c)** With one named example, describe the role of proto-oncogenes and tumour suppressor genes in the control of normal cellular division. [8]

[Total: 20]

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